

Setting Mail server in Ubuntu 22

Initial server setup

Configure server domain name

```
sudo apt -y update; sudo apt -y upgrade
sudo echo 'dev.dsinnovators.com' > /etc/hostname
sudo sed -i 's/^(127.0.0.1\).*$/\1 dev.dsinnovators.com/' /etc/hosts
sudo timedatectl set-timezone Asia/Dhaka
sudo reboot now
```

Setup Mail Transfer Agent

Install required packages

```
sudo apt install -y mailutils
sudo apt install -y postfix
```

Update Postfix configuration

```
sudo dpkg-reconfigure postfix
> Internet Site
> System mail name: dev.dsinnovators.com
> Recipient for root: <enter>
> Other destinations to accept mail: dev.dsinnovators.com, localhost.dsinnovators.com, localhost
> Force synchronous updates: <No>
> Local networks: 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
> Mailbox size limit: 0
> Local address extension character: +
> Internet protocols to use: all
```

Custom configuration

```
sudo vi /etc/postfix/main.cf

# Update
home_mailbox = Maildir/
inet_interfaces = loopback-only
masquerade_domains = dev.dsinnovators.com
```

Reload Postfix

```
sudo systemctl restart postfix
postconf -n
```

Verify server connection

```
telnet smtp.google.com 25
> ehlo smtp.google.com
```

DKIM key setup

Install DKIM packages

```
sudo apt install -y opendkim opendkim-tools
```

Add Postfix user to OpenDKIM group

```
sudo usermod -aG opendkim postfix
```

Setup OpenDKIM key directory

```
sudo mkdir -p /etc/opendkim/keys
sudo chown -R opendkim:opendkim /etc/opendkim
sudo chmod 744 /etc/opendkim/keys
```

Generate DKIM key for website

```
sudo mkdir /etc/opendkim/keys/dev.dsinnovators.com
sudo opendkim-genkey -b 2048 -d dev.dsinnovators.com -D /etc/opendkim/keys/dev.dsinnovators.com -s default -v
sudo chown opendkim:opendkim /etc/opendkim/keys/dev.dsinnovators.com/default.private
```

Update DKIM key in DNS record

```
sudo cat /etc/opendkim/keys/dev.dsinnovators.com/default.txt

# Output
default._domainkey IN TXT ("v=DKIM1; h=sha256; k=rsa; p=Encrypted_key");
```

Verify the DNS record

```
sudo opendkim-testkey -d dev.dsinnovators.com -s default -vvv
```

Update DKIM configuration

```
sudo vi /etc/opendkim.conf

# Update
LogWhy          yes
Mode            sv
SubDomains      no

# Add
AutoRestart     yes
AutoRestartRate 10/1M
Background      yes
DNSTimeout      5
SignatureAlgorithm rsa-sha256

# End of the file
KeyTable        refile:/etc/opendkim/key.table
SigningTable     refile:/etc/opendkim/signing.table
ExternalIgnoreList /etc/opendkim/trusted.hosts
InternalHosts    /etc/opendkim/trusted.hosts
```

Additional DKIM Files

Update Signing table

```
sudo vi /etc/opendkim/signing.table

# Add
*@dev.dsinnovators.com default._domainkey.dev.dsinnovators.com
*/*.dev.dsinnovators.com default._domainkey.dev.dsinnovators.com
```

Update Key table

```
sudo vi /etc/openskim/key.table

# Add
default._domainkey.dev.dsinnovators.com dev.dsinnovators.com:default:/etc/openskim/keys/dev.dsinnovators.com/default.private
```

Update hosts

```
sudo vi /etc/openskim/trusted.hosts

# Add
127.0.0.1
localhost
.dev.dsinnovators.com
```

Restart OpenDKIM

```
sudo systemctl restart openskim
```

Configuring Postfix With OpenDKIM

Update OpenDKIM socket file

```
sudo mkdir /var/spool/postfix/openskim
sudo chown openskim:postfix /var/spool/postfix/openskim
```

Update configuration file

```
sudo vi /etc/openskim.conf

# Update
Socket local:/var/spool/postfix/openskim/openskim.sock
```

Update OpenDKIM configuration

```
sudo vi /etc/default/openskim

# Update
SOCKET="/var/spool/postfix/openskim/openskim.sock"
```

Update Postfix configuration

```
sudo vi /etc/postfix/main.cf

# Add
# Milter configuration
milter_default_action = accept
milter_protocol = 6
smtpd_milters = local:openskim/openskim.sock
# smtpd_milters = local:/var/spool/postfix/openskim/openskim.sock
non_smtpd_milters = $smtpd_milters
```

Restart services

```
sudo systemctl restart openskim
sudo systemctl restart postfix
```

Sending Email

```
echo "Hello, from Rony" | mail -s "This is a test mail" <user>@yahoo.com
```

Thank you.